Business Guesstimation & Forecasting — Structured Notes

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Study sequence

Padhne ka best order: Guesstimation Sessions 1 → 4, then the practice case, then the assignment, then Forecasting. Each session builds on the last, and Forecasting is literally "Stage 3" after estimation.

#	File	Why at this point
1	Session 1 – Introduction to Guesstimation	Base concepts: guess vs guesstimation, 3 levels of estimation
2	Session 2 – Fermi Estimation	The 6-step framework you'll use in every problem
3	Session 3 – Top-Down vs Bottom-Up (slides)	Visual summary of the two approaches
4	Session 3 – Instructor Notes	Same topic in depth, with worked examples
5	Session 4 – Cognitive Biases (Instructor Notes)	Why estimates go wrong and how to debias
6	Practice Case Guesstimation 2	Try it yourself first
7	Practice Case Solution	Then check against the model answer
8	Consulting Assignment (PDF = DOCX, same content)	Apply everything to the graded task
9	Forecasting Session 1 – Intro to Forecasting	Moves from "no data" to "historical data"
10	Forecasting Session 3 – AR and MA	Time-series models

Forecasting Session 2 (naïve, moving averages, Holt, Holt-Winters) is now section 7A, from the instructor's course web page. The Consulting Assignment .docx and .pdf are the same document.

1. Introduction to Guesstimation

Managers cannot wait for perfect information; guesstimation gives a structured, defensible estimate so today's decision can be made.

Definition. Guesstimation = Guess + Estimation. It is arriving at a reasonable estimate by combining logical reasoning, assumptions, decomposition and proxy information when complete data is unavailable. Keywords: *reasonable, logical, incomplete*.

Why managers estimate. Most big decisions have little or no history to lean on.

Decision	Historical data?
New branch	No
New product	No
New country	No
New pricing	Little
New hiring	Partial

Cases discussed: Nestlé Japan (tea-to-coffee market entry) and Bisleri India. Myth: Perfect data → perfect analysis → perfect decision. Reality: decisions are made with partial data.

Case slides are image-only: Nestlé Japan (Nescafé entering a tea-drinking culture) and Bisleri India (bottled-water market shown as ~$360 mn by 2029, Bisleri the largest share, "others" ~45%). Both show managers sizing a market with no historical data. Read the links on slide 10 for the full stories.

Guess vs Guesstimation

Guess	Guesstimation
Random	Structured
Opinion	Logical

Guess	Guesstimation
Hidden assumptions	Explicit assumptions
Cannot defend	Can defend
Cannot improve	Continuously improved

3 levels of estimation

Level	Information available
Guess	Almost none
Guesstimation	Partial
Forecast	Historical data available

Anatomy of guesstimation (with feedback loops back to earlier steps):

```
flowchart LR
  A[Problem] --> B[Break into components]
  B --> C[Estimate components]
  C --> D[Combine estimates]
  D --> E[Validate]
  E --> F[Decision]
  E -.-> B
  F -.-> C
```

Managerial estimation. Guesstimation answers "How much?" (logical guesses, order-of-magnitude, incomplete data). Managerial estimation answers "So what should we do?" Example: estimate of 1,000 customers/day drives decisions on hiring, inventory, seating, working capital, cash and pricing.

From estimate to decision (opening a café): Unknown → estimate customers/day → estimate sales (avg spend × customers) → estimate fixed + variable costs → profit = sales − costs → should we open?

In-class exercises: cups of tea sold inside Jaipuria Noida; a Starbucks at your institute (think customers, coffee demand, revenue, peak hours, seating, staff).

Good estimation is: transparent, logical, defensible, updateable, action-oriented. **Poor estimation sounds like:** "I feel…", "My friend said…", "It should be…".

Think like a manager: not "What is the exact answer?" but "What is a reasonable estimate that helps me make today's decision?" Discussion point: a wrong estimate can still produce a good decision if the logic is sound.

Guesstimation vs Analytics vs Forecasting

Aspect	Guesstimation	Analytics	Forecasting
Purpose	Estimate	Understand	Predict
Data needed	Little or none	Current & historical	Historical time series
Basis	Logic & assumptions	Data analysis	Statistical models
Typical stage	Before launch	During operations	Mature operations
Example	Market size	Why sales fell	Next month's sales
Output	Approximate estimate	Business insights	Future values

Takeaways: managers can't wait for perfect data; guesstimation is structured thinking; every assumption must be explicit; the goal is better decisions, not precision.

2. Fermi Estimation & Assumption-Based Reasoning

Fermi estimation breaks a big unknown into smaller, estimable pieces to reach an order-of-magnitude answer; the reasoning path matters more than the final digit.

Opening case: "How many Blinkit orders daily in Noida?" Class answers ranged 5,000 to 1,000,000 (a 200× spread). Gut feel jumps to a number; managers first identify variables, then estimate each. Structure collapses the range.

Core principle: instead of "What is the answer?", ask "What smaller, simpler questions can I answer to get there?"

Enrico Fermi (1901–1954): Nobel physicist. At the Trinity test he estimated blast strength by dropping paper scraps and watching how far they scattered. Classic question: "How many piano tuners in Chicago?" (population → households → piano owners → tuning frequency → tuner capacity). Now standard in McKinsey, BCG, Bain and Google interviews.

Characteristics of Fermi estimation: logical & systematic, transparent & auditable, flexible & updatable, ideal without historical data. Uses: market sizing, demand for new launches, workforce/capacity planning, baseline budgets.

Characteristics of Fermi problems: data scarcity, ambiguity (feels too big to start), assumption-driven, and transparency of reasoning is prioritised over precision.

Why managers need it: day-1 trade-offs (inventory, headcount, warehouse sq. ft.) must be made before a single sale. Without structure you get analysis paralysis or reckless guessing.

The 6-step Fermi framework

```
flowchart LR
  A[1 Define the unknown] --> B[2 Decompose]
  B --> C[3 Assume & assign]
  C --> D[4 Arithmetic]
  D --> E[5 Validate]
  E --> F[6 Refine iteratively]
  F -.-> C
```

1. **Define the unknown** — make the target metric exact. Vague: "Figure out demand". Specific: "Expected daily customer orders for the new dark store in Noida". Also define warehouse size, rider count, inventory, marketing budget with the same precision.
2. **Decompose** — break into linear, estimable steps. Blinkit tree: Total households → Smartphone-enabled → Q-commerce active → Orders/month → Daily orders. Benefits: isolates where errors sit, enables sensitivity testing, easy to explain to a board.
3. **Assume & assign** — an assumption is a reasonable, defensible statement accepted as true when evidence is absent. Good ones are explicit, evidence-linked, malleable, consistent, verifiable. Use proxy variables where direct data is missing.
4. **Arithmetic** — multiply/divide the chain.
5. **Validate** — sanity-check against physical/operational limits; run scenarios.
6. **Refine iteratively** — Estimate → Validate → Learn (post-launch data) → Improve (replace proxies with real inputs), repeat.

Weak vs strong assumptions

Dimension	Weak (intuitive)	Strong (evidence-linked)
Market scaling	"A ton of households will order online"	"~30% of target tier-1 households use quick commerce"
Ordering behaviour	"Customers will order very frequently"	"~5 orders/month per active user, from regional cohort data"
Operational limits	"Store will handle all traffic effortlessly"	"Sorting staff capped at 120 orders/hour"

5 types of assumptions: Demographic (population, household size, density) · Behavioural (order frequency, basket size, loyalty) · Economic (income, price sensitivity) · Operational (delivery radius, sorting speed, capacity) · Technological (app adoption, internet, digital payments).

Proxy variables: substitute an unknown with a correlated, observable index. E.g. household inflow via census maps within 3 km; discretionary spend via housing prices and premium smartphone share; digital readiness via digital-wallet penetration.

Worked example — daily orders for one Noida dark store

Step	Value
Catchment households	50,000
× 40% smartphone penetration	20,000
× 20% Q-commerce adoption	4,000 active customers
× 4 orders/month	16,000 orders/month
÷ 30 days	~533 orders/day

Follow-up questions it triggers: can the store physically handle 533/day? At 20 deliveries per rider per shift, how many riders? What inventory does it imply?

Sensitivity (Step 5)

Scenario	Adoption	Orders/day	Action
Worst	15%	400	Cut staffing, reduce marketing spend

Scenario	Adoption	Orders/day	Action
Expected	20%	533	Keep planned ratios and capex
Best	30%	800	Backup rider contracts, overflow sorting zones

Common mistakes: jumping to a gut answer; unrealistic assumptions (100% adoption, daily purchases by whole city); omitting gatekeeping variables (income, competitors, internet); burying assumptions in a black box; skipping the sanity check.

Estimation → analytics ladder: Stage 1 Fermi estimation (no history, logic + proxies) → Stage 2 operational tuning (pilot data) → Stage 3 robust forecasting (deep history, statistics/ML). Every forecasting model started life as a Fermi estimate.

3. Top-Down vs Bottom-Up Estimation

Both approaches estimate the same quantity from opposite ends: top-down filters a big market down (demand side); bottom-up multiplies a small operating unit up (supply side). Use both and compare — that is triangulation.

Learning outcomes: apply both approaches, distinguish demand-driven vs supply-driven estimation, choose the right framework, build logical assumptions.

Top-down (demand-side funnel)

Starts with a large aggregate (population, market size, industry demand) and narrows it with logical filters.

5 stages: 1) identify total market/population → 2) divide into relevant segments → 3) apply logical assumptions at each stage → 4) eliminate irrelevant groups → 5) arrive at final estimate.

Illustration (quick-commerce orders in Delhi): Delhi population → adults → internet users → Q-commerce users → avg orders per user → daily orders.

- **Advantages:** easy to understand; works when operational data is unavailable; good for market potential, new businesses and emerging industries.
- **Limitations:** heavy reliance on assumptions; an error at the top of the funnel skews everything below; tends to *overestimate* demand.

Bottom-up (supply-side foundation)

Starts with the smallest operational unit and aggregates upward. Estimates how much business activity capacity can generate, not how many people want the product.

5 stages: 1) identify smallest measurable unit → 2) estimate output of one unit → 3) estimate number of such units → 4) multiply output per unit × number of units → 5) final estimate.

Illustration (Blinkit orders in Delhi): one dark store → orders/hour → × operating hours → orders per store per day → × number of dark stores → daily orders.

- **Advantages:** grounded in operating reality; more realistic capacity estimates; ideal for production, logistics and operations planning.
- **Limitations:** needs internal knowledge of the unit; hard if the business is a black box; may *underestimate* demand if capacity is constrained.

Comparison matrix

Basis	Top-Down	Bottom-Up
Starting point	Total market or population	Smallest operational unit
Primary focus	Customer demand	Supply / capacity
Direction	Large → small	Small → large
Information required	Market assumptions	Operational assumptions
Best used for	Market sizing, new opportunities	Revenue, production, logistics, capacity
Examples	OTT subscribers in India, EV market, demand for online education	Plant annual output, food deliveries by one company, daily ATM transactions
Common users	Consultants, strategy teams	Operations managers, supply-chain analysts
Error tendency	Overestimates	Underestimates

Choosing: use top-down when sizing a market, estimating customer demand, evaluating a new opportunity, or when operational info is limited. Use bottom-up when estimating capacity, company revenue or operational output, and when internal info is available.

Triangulation

Experienced analysts compute the same number both ways. If the two are close → confidence rises (Path A). If they differ a lot → assumptions are flawed; revisit them (Path B). This is the gold standard in consulting, market research and financial analysis.

Worked example — Blinkit daily orders in Delhi: top-down (population → adults → internet users → Q-commerce users → × orders per customer) vs bottom-up (orders per dark store per hour → × operating hours → × number of dark stores). Similar results = realistic assumptions.

Key line: "In business guesstimation, the quality of assumptions is more important than the precision of calculations." Neither approach is superior; the problem and the available information decide.

4. Estimation under Uncertainty & Cognitive Biases

The biggest source of estimation error is poor assumptions, not wrong arithmetic, and cognitive biases corrupt assumptions before any calculation starts.

Estimation under uncertainty: making informed judgments when information is incomplete. Forecasting leans on statistical models and history; guesstimation leans on structured thinking. Goal: *reduce* uncertainty through logical assumptions, not eliminate it.

Sources of uncertainty: incomplete market info, rapid tech change, customer behaviour, competition, government regulation, economic conditions, random events.

Example — premium coffee shop near a university

Version	Chain	Customers/day	Revenue/day (₹180 bill)
Initial	8,000 students × 20% buy	1,600	₹2,88,000
Refined	8,000 × 40% coffee drinkers × 25% premium × 20% actually visit	160	₹28,800

Questions that forced the refinement: do all students drink coffee? will they pay premium? how many already buy from competitors? what % visit daily? Better assumptions cut the estimate 10×.

Cognitive bias = a systematic error in thinking that causes irrational or inaccurate judgments.

The five biases

Bias	What it is	Business case	Fix — ask yourself
Anchoring	Over-relying on the first number heard	"I heard Company X revenue is ₹500 cr" → everyone guesses ₹400–700 cr	Ignore outside numbers; build from drivers: 250 stores × ₹6 lakh/day × 365 ≈ ₹547 cr
Availability	Overweighting recent, memorable or much-discussed events	IPL-season food-delivery ads → students guess 2 million daily orders in Delhi	"Is this common or just memorable?" Build from population → households → users → orders
Confirmation	Seeking only evidence that supports your belief	EV entrepreneur ignores charging gaps, high prices, rural adoption → demand overestimated	"What evidence proves me wrong?" Search for contradictory data
Overconfidence	Believing your estimate is more accurate than it is	"One dark store = 600 orders/day" with no evidence	"What if I'm wrong?" Use a range: min 400, most likely 550, max 700
Representativeness	Treating one example as the whole population	Crowded Starbucks at Delhi Airport → "every Starbucks serves 800/day"	"Is this example representative?" Airport, office, mall and residential stores differ

Scenario analysis

Professionals give three scenarios, not one number.

Scenario	Daily customers	Revenue (₹200/customer)
Worst	120	₹24,000
Most likely	180	₹36,000
Best	250	₹50,000

Assumption refinement: ask "How good are my assumptions?" instead of "What is the answer?". Example: "30% of Delhi uses Blinkit every day" → revised to ~30% use it at all, and an active customer places ~0.12 orders/day.

Case — daily food-delivery orders in Noida

Step	Working
1. Initial	10,00,000 population × 25% order daily = 2,50,000/day
2. Bias found	Availability bias (saw crowded restaurants)
3. Refined	10,00,000 × 70% adults × 60% working × 35% delivery users × 0.18 orders/day ≈ 26,460/day

De-biasing checklist (any "No" → revisit assumptions):

1. Did I avoid anchoring on someone else's number?
2. Am I free of recent-event influence?
3. Did I look at contradictory evidence?
4. Am I not generalising from one example?
5. Did I use ranges for uncertainty?
6. Can I justify every assumption?

Takeaway: a good analyst doesn't hit the "right" answer first time; they iteratively question and validate every assumption.

5. Practice Case — K-Pop Concert in Delhi (Case 5)

Answer: maximum ticket revenue ≈ ₹12.5 crore (25,000 seats × ₹5,000), because demand (~1.8 lakh) far exceeds venue capacity. The bottleneck is capacity, not market size.

Given: Delhi-NCR population 30 million; 20% aged 15–30; 15% of them follow K-Pop; 20% of followers would consider a live concert; avg ticket ₹5,000; venue 25,000; one visit per customer.

Step 1 — Clarify: "How much ticket revenue can a one-night concert realistically generate?" (decides artist fee, venue, marketing commitment).

Step 2 — Classify: Guesstimation. Some data is known, but actual ticket demand is not, and there is no time series to forecast.

Step 3 — Fermi tree (top-down):

Stage	Working	Result
Population	Given	3,00,00,000
Age 15–30	× 20%	60,00,000
K-Pop followers	× 15%	9,00,000
Would consider attending	× 20%	1,80,000
Venue cap	min(1,80,000, 25,000)	25,000
Revenue	25,000 × ₹5,000	₹12.5 crore

Step 4 — TAM / SAM / SOM

Term	Meaning	Here
TAM	Everyone with a plausible desire for the product (not the whole city)	1,80,000 potential attendees
SAM	Portion the organiser can actually serve	25,000 seats (venue limit)
SOM	Portion realistically captured	25,000 tickets, since demand ≫ capacity

Step 5 — Revenue, not profit. Artist fee, venue, marketing, security, production, taxes and ticketing fees are not given, so profit cannot be computed. "Don't calculate what the data doesn't support."

Step 6 — Capacity insight: only 25,000 / 1,80,000 ≈ 13.9% of potential demand can be served.

Step 7 — Sanity check (internal): 25,000 is only ~2.8% of 9 lakh followers and ~0.083% of Delhi-NCR's population. Passes an order-of-magnitude check.

Step 8 — What could make it wrong:

1. K-Pop penetration (15% → 10% cuts demand a lot).
2. Concert conversion — "consider attending" ≠ "buys a ticket" (biggest weakness).
3. Artist attractiveness — superstar vs mid-tier have very different pull.
4. Ticket price — real mix could be ₹2,500 to ₹10,000+ and VIP packages.
5. Saleable seats < 25,000 (production areas, VIP, blocked seats, crew, safety).

Step 9 — Sensitivity: potential attendees = 60 lakh × follow% × conversion%; tickets = min(attendees, 25,000); revenue = tickets × ₹5,000.

Follow % ↓ / Conversion % →	10%	20%	30%
10%	60,000	1,20,000	1,80,000
15%	90,000	1,80,000	2,70,000
20%	1,20,000	2,40,000	3,60,000

Every cell is above 25,000, so revenue stays ₹12.5 crore in all nine scenarios. Lesson: once demand far exceeds capacity, a better demand estimate doesn't raise revenue; price and saleable seats matter more.

Step 10 — Recommendation (model answer): Commercially attractive on ticket revenue (~₹12.5 cr); capacity is the main constraint. Before committing, validate demand with artist-specific fan data and pre-registration, and refine the pricing strategy. Caveat: profitability cannot be concluded without costs.

11 interview questions to practise: classification · Fermi breakdown · top-down/bottom-up choice · TAM/SAM/SOM · capacity constraint · extra assumptions · step-by-step calculation · sanity check · model weaknesses · most sensitive assumption · business recommendation.

6. Consulting Assignment — Project NutriBox

The assignment asks one question: should NutriBox open a delivery kitchen in Noida Sector 62? You build an Excel business case and end with GO / NO-GO / GO WITH CONDITIONS. It uses exactly the K-Pop case logic plus costs and break-even.

Client data

Area	Input	Value
Market	Office employees in catchment	1,20,000
Market	Buy lunch outside/online	60%
Market	Of these, interested in healthy meals	35%
Market	Realistic first-year market share	8%
Pricing	Avg meal price	₹180
Demand	Working days/month (customers)	22
Demand	Orders	1 meal/customer/day
Capacity	Kitchen operating days/month	26
Capacity	Kitchen capacity	500 meals/day
Capacity	Riders × meals per rider/day	20 × 25
Costs	Variable cost/meal	₹100
Costs	Monthly fixed operating cost	₹8,00,000
Costs	Initial kitchen investment	₹30,00,000

How each question maps to the course

Q	What to compute	Method from notes
1 Market sizing	TAM = catchment employees; SAM = TAM × 60% × 35%; SOM = SAM × 8%	Top-down funnel (Session 3), TAM/SAM/SOM (Case 5)

Q	What to compute	Method from notes
2 Revenue	Customers = SOM; meals/day = customers × 1; monthly = × 22; annual = × 12; revenue = meals × ₹180	Fermi arithmetic (Session 2)
3 Capacity	Kitchen = 500/day; delivery = riders × meals/rider; effective = min(kitchen, delivery); surplus/shortfall = effective − demand	Bottom-up (Session 3), bottleneck idea (Case 5)
4 Profitability	Variable cost = meals × ₹100; fixed = ₹8 lakh × 12; operating profit = revenue − variable − fixed; break-even meals = fixed ÷ (price − variable cost)	Contribution margin
5 Sensitivity	Share 5/8/11%, price ₹160/180/200, variable cost ₹90/100/115 → revenue and profit each	Scenario analysis (Sessions 2 & 4)
6 Critical assumptions	Top 3 drivers, most uncertain one, extra data needed	Assumption refinement, bias checklist (Session 4)
7 Recommendation	3 key numbers, 3 assumptions, 1 risk, 1 action	Model answer structure from Case 5

Watch-outs:

- Meals served = min(demand, effective capacity). If demand exceeds capacity, revenue is capped, exactly like the concert venue.
- Customers order on 22 working days but the kitchen runs 26; use 22 for demand-driven meals and state that assumption.
- Use break-even per month and per day to compare with capacity; mention payback on the ₹30 lakh investment if profit is positive.
- Say "operating profit" (no tax, depreciation or interest given) and don't invent numbers the brief doesn't support.
- Run the de-biasing checklist on the 60% / 35% / 8% inputs before finalising.

7. Forecasting Session 1 — Introduction to Business Forecasting

Business forecasting is the systematic process of using historical data, current information and analytical models to estimate future business outcomes; it reduces uncertainty but always carries error.

Learning outcomes: explain forecasting and its relevance; differentiate forecasting from prediction and estimation; explain why it matters; recognise forecasting horizons.

Old "forecasting": horoscopes, crystal ball, tea leaves, numerology, palmistry — no data, no model.

Why forecast? Reduces uncertainty, helps managers gauge the situation, helps companies prepare. Examples: RBI forecasts inflation (economics); airlines forecast passenger traffic (operations); analysts forecast earnings (finance); HR forecasts staff needs.

Applications: business planning & operations (demand, inventory, resource allocation, production scheduling); economic & financial planning (inflation, interest rates, currency, capital budgeting); environment & climate (weather, disasters, sustainability); supply chain & logistics (shipping routes, transport capacity, warehousing, demand spikes). Starbucks example: morning coffee demand → milk procurement → staff scheduling → inventory → revenue.

Guess vs guesstimate vs estimation vs prediction vs forecasting

Concept	Focus	Time orientation	Data rigour	Example
Guess	Pure conjecture	Any	Minimal/none	Beans in a jar without counting
Guesstimate	Order-of-magnitude approximation	Any	Low (Fermi)	Napkin market size from population density
Estimation	Unknown parameters or current/past values ($\hat{\theta}$)	Present/static	High (statistical)	Mean income, project cost

Concept	Focus	Time orientation	Data rigour	Example
Prediction	Unobserved outcome (ŷ) given inputs	Future or out-of-sample	High (algorithm)	Will this loan applicant default?
Forecasting	Future time-series path (ŷ at t+h)	Explicitly future	High (time-series, econometrics)	Next quarter's revenue from past sales

Quick test: "Starbucks customers today?" = guesstimation; "customers next month?" = forecasting; "Will India win the 2027 ODI World Cup?" = prediction.

Forecast error: every forecast has error; measured later with MAE, MAPE, RMSE.

Forecasting horizons

Horizon	Span	Examples
Short	Hours, days, weeks	Daily sales, Uber demand, weather
Medium	Months	Quarterly sales, budget planning
Long	Years	Factory expansion, new market entry, GDP growth

Accuracy falls as the horizon lengthens.

Types of data: cross-sectional (many units at one time — used in regression), time series (one unit over time), panel (both). Regression so far used cross-sectional data.

Data type	Meaning	Example
Time series	One subject over time	Monthly sales of ONE store; yearly GDP of one country
Cross-sectional	Many subjects at ONE point in time	Sales of 100 stores in Dec 2024; stock prices of 10 companies today
Panel	Many subjects over time	Yearly sales of 100 stores, 2020–2024

Case — UltraBuild Cement Ltd, forecast 2025 Q1 sales (for production, inventory, procurement, capacity, revenue budgeting):

$$\text{Sales} = \beta_0 + \beta_1\,\text{Advertising} + \beta_2\,\text{Housing Starts} + \beta_3\,\text{Price} + \beta_4\,\text{Rainfall} + \beta_5\,\text{GDP Growth} + \varepsilon$$

Variable	Unit	Expected sign
Advertising	₹ lakh	+
Housing starts	'000 units	+
Price	₹/bag	−
Rainfall	mm/quarter	− (construction slows)
GDP growth	%	+

Yes, you can forecast 2025 Q1 with regression — but only for specific (assumed) values of the independent variables. The open question "Can past sales predict future sales?" leads to time-series models.

7A. Forecasting Session 2 — Classical Forecasting Techniques

Source: Dr. Sankalp Naik's course page, Classical Forecasting Techniques. Every forecast tries to separate signal from noise. Start with naïve benchmarks, smooth out noise with moving averages, then add trend (Holt) and seasonality (Holt-Winters) only when the data shows them.

Data Generating Process (DGP)

A DGP is the statistical mechanism that actually produces the data. We never observe the true DGP; a model is a simplified stand-in for it.

$$\text{Observed data} = \text{Signal (systematic pattern)} + \text{Noise (random variation)}$$

Illustration — study hours vs exam score (10 students):

Hours	1.5	2.0	3.5	4.0	5.5	6.0	7.5	8.0	9.5	10.0
Score	43	47	56	59	68	70	78	81	87	89

OLS result: Score = 36.74 + 5.40 × Study Hours; R^2 = 0.994; F = 1,423. The intercept (36.74) is the predicted score at 0 hours. The slope says each extra hour adds ~5.4 marks. The fitted line is the *signal*; residuals (actual − predicted) estimate the *noise*.

Key idea: forecasting = identify the systematic part of the DGP and project it forward. In time series the signal shows up as **level, trend, seasonality or cycles**.

Naïve methods (benchmarks)

Simple, no modelling, used as the baseline every other model must beat.

Method	Rule	Use when
Naïve	F(t) = Y(t−1): next value = last value	Random-walk data, e.g. stock prices
Seasonal naïve	F(t) = Y(t−s): same season last cycle (s = 4 for quarters, 12 for months)	Strong repeating seasonality

Naïve example — stock price (₹): Jan 1,842.50 · Feb 1,875.25 · Mar 1,821.75 · Apr 1,906.50 · May 1,958.25 · Jun 1,932.75 · Jul 2,014.50 · Aug 2,067.25 · Sep 2,041.50 → **October forecast = ₹2,041.50**.

Seasonal naïve example — quarterly sales (₹ lakh):

Quarter	2024	2025	2026 forecast
Q1	100	98	98
Q2	120	122	122
Q3	105	103	103
Q4	145	148	148

Smoothing: moving averages

Smoothing averages past values to filter out noise; best for series that hover around a level.

Simple Moving Average (SMA) over k periods:

$$F_t = \frac{Y_{t-1} + Y_{t-2} + \dots + Y_{t-k}}{k}$$

Dimension	Short window (e.g. 3)	Long window (e.g. 10)
Observations used	Fewer	More
Responsiveness	High	Low
Noise sensitivity	High	Low
Smoothing	Minimal	Maximum
Lag	Lower	Higher
Best when	Recent changes matter	Pattern is stable

Trade-off: short window = responsive but noisy; long window = smooth but slow. Case used: quarterly automobile sales (80 quarters; sales plus GDP, interest rate, ad spend, fuel price, consumer confidence). MA(3) follows short-term moves; MA(10) shows the broad direction.

Weighted Moving Average (WMA) gives more weight to recent data; weights must sum to 1.

$$F_t = w_1 Y_{t-1} + w_2 Y_{t-2} + w_3 Y_{t-3}, \quad w_1 + w_2 + w_3 = 1$$

Example weights: 0.5 / 0.3 / 0.2 (slides) or 0.6 / 0.3 / 0.1 (Python demo, most recent = 0.6). Analogy: a 10-year performance review that puts 80% weight on the last two years.

SMA vs WMA: SMA treats all observations equally; WMA reacts faster at turning points, has less lag, and lets you choose how fast memory decays — while still smoothing.

Holt's method (double exponential smoothing) — level + trend

For data with a sustained upward or downward direction. It answers "Where is the series now, and where is it heading?"

$$L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta (L_t - L_{t-1}) + (1-\beta) T_{t-1}$$

$$F_{t+h} = L_t + h \, T_t$$

α = level smoothing (0–1), β = trend smoothing (0–1), h = periods ahead. Higher α/β = more weight on recent data. Python: `ExponentialSmoothing(series, trend="add")`. Typical uses: revenue with steady growth, stock prices.

Holt-Winters (triple exponential smoothing) — level + trend + seasonality

For data with both drift and a repeating cycle (s = season length, e.g. 4 for quarters).

$$L_t = \alpha (Y_t - S_{t-s}) + (1-\alpha)(L_{t-1} + T_{t-1})$$

$$T_t = \beta (L_t - L_{t-1}) + (1-\beta) T_{t-1}$$

$$S_t = \gamma (Y_t - L_t) + (1-\gamma) S_{t-s}$$

$$\text{Additive: } F_{t+h} = L_t + h\,T_t + S_{t+h-s} \qquad \text{Multiplicative: } F_{t+h} = (L_t + h\,T_t)\, S_{t+h-s}$$

γ = seasonal smoothing (0–1). **Additive** when seasonal swings stay the same size; **multiplicative** when swings grow with the level. Python: `ExponentialSmoothing(series, trend="add", seasonal="add", seasonal_periods=4)`. Uses: retail sales, electricity demand, tourism.

Progression chart — which model when

Model	Components	Best for
Naïve / seasonal naïve	Last value / last season	Benchmark
SMA / WMA	Average level	Noisy series with no trend
Simple exponential smoothing	Level	Stable, non-trending series
Holt (double)	Level + trend	Sustained upward/downward movement
Holt-Winters (triple)	Level + trend + seasonality	Drift plus repeating cycle

Rule: start simple and add a component only when the data shows it.

Simple exponential smoothing appears on the site only in this chart. Its standard formula (my addition, not from the site) is $F(t+1) = \alpha Y(t) + (1-\alpha)F(t)$.

Worked hand example (my practice illustration, not from the site)

The AR/MA warm-up series: 100, 105, 108, 112, 115 → forecast period 6 with each method.

Method	Working	Forecast for t = 6
Naïve	Last value	115.00
SMA(3)	(108 + 112 + 115) ÷ 3	111.67
WMA(3), weights 0.5/0.3/0.2	0.5×115 + 0.3×112 + 0.2×108	112.70
Holt, α = β = 0.5	$L_5 + T_5$ (see below)	119.59

Holt steps (start $L_1 = 100$, $T_1 = Y_2 - Y_1 = 5$):

t	Y	L(t)	T(t)
1	100	100.00	5.00
2	105	105.00	5.00
3	108	109.00	4.50
4	112	112.75	4.13
5	115	115.94	3.66

$F_6 = 115.94 + 3.66 = 119.59$; $F_7 = 115.94 + 2 \times 3.66 = 123.25$. The series has a clear uptrend, so SMA/WMA lag behind; Holt carries the trend forward. That's why model choice depends on the pattern.

8. Forecasting Session 3 — Time Series: AR and MA Models

AR models use the series' own past *values* to forecast; MA models use past *shocks* (forecast errors). AR(1) captures persistence of values, MA(1) persistence of shocks.

Recap of earlier sessions (Session 2 file not shared): time-series decomposition; classical models — Naïve and Seasonal Naïve; smoothing — Moving Average and Weighted Moving Average; trend & seasonality — Holt and Holt-Winters.

Regression vs time series

Regression	Time series
Relationship between Y and explanatory variables	Behaviour of one variable over time
"What factors explain Y?"	"How does Y evolve over time?"
Observations often treated as independent	Order matters; observations depend on earlier ones
Explanatory variables are external to Y	Past Y itself informs future Y
Sales = f(price, advertising, income)	Sales today = f(sales yesterday, trend, seasonality, shocks)
Autocorrelation is a problem to fix	Autocorrelation is the information to model
Used for explanation/prediction	Used for forecasting and temporal dynamics

Serial correlation (autocorrelation): the statistical relationship between a variable's current value and its past values across periods. Plots of errors against time or against the previous error show cyclical, rising or falling patterns when it's present.

Time-series decomposition (recap slide): actual demand = trend (long-run upward/downward line) + seasonal peaks (repeat each year) + random fluctuation around them, compared against the flat long-run average. Trend patterns: uptrend (higher highs, higher lows), sideways trend (trading range), downtrend (lower highs, lower lows).

Warm-up: 100 → 105 → 108 → 112 → 115 → ? The next value clearly depends on recent values — that dependence is what AR models.

AR(1) — autoregressive

Idea: the past of Y helps predict the future of Y.

```
Y_t = b_0 + b_1\,Y_{t-1}
```

b_1 (also written φ) = **persistence**. Interpretation: if yesterday's value is 1 unit higher, today's expected value is b_1 units higher, all else equal.

MA(1) — moving average

Idea: what matters may not be yesterday's value but yesterday's *unexpected event*. If expected was 100 and actual was 120, the extra 20 is a **shock**; MA models capture persistence + shock.

```
Y_t = c + \varepsilon_t + \theta\,\varepsilon_{t-1}
```

(The formula is marked "not for exam" in the slides; know the idea.)

AR vs MA summary

	AR(1)	MA(1)
What matters	Past value	Past shock
Input	Y(t−1)	ε(t−1)
Parameter	φ = persistence of values	θ = persistence of shocks

Don't confuse MA(1) here with the "moving average" smoothing method from earlier sessions: smoothing averages past values; the MA model uses past errors.

9. Quick Revision Sheet

One-page recall for the night before the exam.

Topic	Remember this
Guesstimation	Guess + estimation: logic + explicit assumptions + decomposition + proxies, with incomplete data
3 levels	Guess (no info) → Guesstimation (partial) → Forecast (historical data)
Anatomy	Problem → break into components → estimate → combine → validate → decision
Good estimate	Transparent, logical, defensible, updateable, action-oriented
Fermi 6 steps	Define → Decompose → Assume & assign → Arithmetic → Validate → Refine

Topic	Remember this
Assumption types	Demographic, behavioural, economic, operational, technological
Good assumption	Explicit, evidence-linked, malleable, consistent, verifiable
Top-down	Market → filters; demand-side; for market sizing; tends to overestimate
Bottom-up	Unit × number of units; supply-side; for capacity/revenue; tends to underestimate
Triangulation	Do both; close = confidence, far apart = fix assumptions
TAM / SAM / SOM	Total interested → what you can serve → what you'll realistically capture
Capacity rule	Sold = min(demand, capacity); find the bottleneck
5 biases	Anchoring, availability, confirmation, overconfidence, representativeness
Scenarios	Always give worst / most likely / best, not one number
Forecasting	Historical data + models → future outcomes; always has error (MAE, MAPE, RMSE)
Horizons	Short (hours–weeks), medium (months), long (years); accuracy drops with horizon
Regression vs TS	"What explains Y?" vs "How does Y evolve?"; autocorrelation is a problem vs the signal
AR(1)	Past value predicts; φ = persistence of values
MA(1)	Past shock predicts; θ = persistence of shocks

Formulas

```
\text{Top-down: } N = \text{Population} \times f_1 \times f_2 \times \dots \times \text{frequency}
```

$$\text{Bottom-up: } N = \text{Output per unit} \times \text{Number of units}$$

$$\text{Break-even units} = \frac{\text{Fixed cost}}{\text{Price} - \text{Variable cost per unit}}$$

Numbers used in class examples: Blinkit Noida store ≈ 533 orders/day (range 400–800); K-Pop concert ≈ ₹12.5 cr (capacity-bound); coffee shop 1,600 → 160 customers/day after refining; Noida food delivery 2.5 lakh → ≈ 26,460 orders/day after de-biasing.